README.md for the Letter Picking Example

LETTER GUESSING GAME - JAVASCRIPT LOGIC SEQUENCE

Overview

This game demonstrates basic JavaScript control flow concepts for an Intro to JavaScript course. The game generates a random letter between A and Z and prompts the user to guess it.

Logic Flow

1. **Initialization** (index.html:69)

    - Generate random target letter: String.fromCharCode(65 + Math.floor(Math.random() * 26))
    - Initialize guessCount = 0
    - Cache DOM elements: messageDiv, guessCountSpan, guessBtn, guessInput

1. **Event Setup** (index.html:76)

    - Add click event listener to the “Submit Guess” button

1. **Per-Guess Processing** (on button click): a. Get user input and convert to lowercase: guessInput.value.toLowerCase() b. Increment guessCount and update display c. **Validate input**: If not a single letter a-z, show error message and exit d. **Convert to numeric equivalents**:

    - Guess: guess.charCodeAt(0) - 96 (a=1, b=2, ..., z=26)
    - Target: targetLetter.toLowerCase().charCodeAt(0) - 96 e. **Compare guess to target**:
    - If guessNum > targetNum: Display “Too high! Try again.” (red background)
    - If guessNum < targetNum: Display “Too low! Try again.” (green background)

- If guessNum === targetNum:
    - Display “Congratulations! You guessed the letter in X guesses!”
    - Turn message green with centered text
- Disable the guess button f. Clear input field and focus back on it

Key JavaScript Concepts Demonstrated

- **Random letter generation**: String.fromCharCode(65 + Math.floor(Math.random() * 26))
- **String-to-number conversion**: charCodeAt(0) - 96 to map A-Z to 1-26
- **Input normalization**: toLowerCase() for case-insensitive handling
- **Comparison operators**: >, <, ===
- **Conditional logic**: if / else if / else chain
- **DOM manipulation**: textContent, className, disabled
- **Event handling**: addEventListener('click', ...)
- **Variable tracking**: Counter variable (guessCount) persistence

Student Takeaways

- How to generate a random letter within the A-Z range
- Converting characters to numeric values for comparison
- Structuring multi-branch conditional logic
- Updating the webpage dynamically without refresh
- Tracking state across multiple user interactions